

Jin Kweon, MSc

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SUMMARY

Bioinformatician with cross-domain expertise spanning clinical (i.e. REDCap) and -omics data across oncology, immunology, and neurodegeneration in both industry and academic settings. Builds scalable pipelines, web-based tools, and published statistical deep learning models (ISMB, IEEE, ASHG) in R and Python.

EDUCATION

McGill University

MSc, Experimental Medicine – GPA:4.00

Thesis: “DTractor: Advancing cell type deconvolution of spatial transcriptomics with deep neural network, transfer learning and matrix factorization”

Montreal, QC

Jan 2023 – Dec 2024

McGill University (Military service: 2019 – 2021)

PhD Candidate, Quantitative Life Science Withdrawn

Montreal, QC

Sep 2018 – Dec 2021

University of California, Berkeley

Bachelor of Arts, Statistics (Focus: Mathematics / Economics) – Major GPA:4.00

Berkeley, CA

Aug 2016 – May 2018

- **Relevant Coursework:** Applied Machine Learning, Structure and Interpretation of Computer Programs, Multivariate Statistics, Linear Algebra, Linear Modeling, Statistical Bioinformatics, Genomics for drug discovery and development

WORK EXPERIENCE

Copoly.ai and Genentech

Computational Biologist

Toronto, ON

May 2025 – Present

- Developed novel statistical framework detecting disease-associated co-abundance at granular, single-cell resolution through differential correlation between cellular neighborhoods, advancing beyond CNA and Milo methods that test neighborhoods independently for differential abundance, missing interaction patterns (accepted ASHG).
- Designed systematic interaction testing across cellular neighborhoods with genetics and spatial integration, revealing coordinated disease responses in AMD, IBD, AD, IPF, and SLE, including enterocyte-immune co-abundance signatures in Crohn's disease.
- Implemented computationally optimized statistical inference pipeline using FastQTL beta approximation and calibrated permutation testing with superior FDR control, enabling robust hypothesis testing across millions of neighborhood pairs.
- Pioneered compartment-based normalization with log-transformed abundance modeling to eliminate compositional bias, ensuring biologically meaningful comparisons across samples with heterogeneous cellular compositions.
- Profiled iPSC-derived RPE differentiation via single-cell annotation, DE, and GSEA, comparing long vs. short protocols to benchmark maturation speed and efficiency while flagging contaminating cell populations for cell therapy quality control.

Montreal General Hospital

Data Analyst

Montreal, QC

Feb 2025 – May 2026

- Developed HuggingFace-based FACT scoring tools and pioneered LLM-based missing data imputation for oncology patient-reported outcomes data (accepted IEEE EMBC), benchmarking 12 methods—from deep learning (VAE/DAE/LSTM) to transformers (Gemma, TimesFM) to ensemble—across cross-sectional and longitudinal metrics.
- Pioneered ESDE 2025 award-winning methodology integrating multiple clinical databases (REDCap, TSQIC, ACCESS) establishing preoperative emotional distress as predictor of prolonged hospitalization and increased readmissions in esophageal cancer patient.
- Revealed in multivariable models that preoperative symptom burden predicted quality-of-life improvement one year post-esophagectomy, with smoking showing a possible trend, while tumor characteristics, surgical approach, and neoadjuvant treatment showed no association; female sex independently predicted reduced improvement.
- Led multi-year analysis (228 patients) demonstrating textbook outcome achievement as both a surgical quality benchmark and prognostic marker, correlating with significantly improved disease-free survival and favorable overall survival in esophageal cancer patients.

Douglas Research Center

Bioinformatician

Montreal, QC

Mar 2025 – May 2025

- Boosted predictive accuracy from 55% to 95% by developing upgraded Python algorithms for meffil and ewaff R packages to detect differentially methylated positions by better leveraging non-linear relationships in the model between CpG methylation sites and phenotype variables.
- Engineered novel spline model methodology improving upon Enmix combp and bumhunter packages, successfully identifying previously undetectable sex-specific methylation regions and site changes across tissues and age.
- Created comprehensive model framework to analyze both sex-dependent and sex-independent methylation effects while ensuring robust evaluation of tissue-specific or tissue-consistent proliferation patterns over age.

Research on developing cell type deconvolution algorithms

Montreal, QC

Computational Biologist (spatial transcriptomics project)

Jan 2023 – Dec 2024

- Implemented the *DTractor* software that demonstrated higher accuracy, resolution, robustness, efficacy, and scalability on two simulated datasets and three real datasets, outperforming eight other popular and commonly used deconvolution methods (accepted ISMB MLCBSB).
- Developed a method that uses deep neural network training on sc/snRNA-seq and spatial transcriptomics (ST) data, incorporating transfer learning from scRNA-seq reference data to ST data, and conducting iterative matrix factorizations.
- Decomposed ST at cell-type resolutions to gain insights into spatial organization in pancreatic adenocarcinoma, uncovering tumor microenvironment interactions and offering rare capability to estimate cell counts per location.

Research on developing AI drug design-related algorithms

Montreal, QC

Research Analyst (single cell project)

Sep 2021 – Nov 2023

- Introduced *scBeacon*, an innovative framework built upon a VQ-VAE framework, deep contrastive siamese network, and a greedy iterative strategy, to effectively pinpoint differential genes and identify the same cell population across different biological conditions forming cluster pairs.
- Advanced biomarker discovery by liberating differential gene analysis from manual annotation constraints, enabling *scBeacon* to identify critical cluster pairs and potential therapeutic targets in COVID-19 datasets.
- Demonstrated therapeutic response by mapping cytarabine-induced cellular adaptations using optimal embedding vector quantization and probabilistic models, revealing elusive drug resistance pathways in treatment populations.

Adecco and Apple

Singapore

Apple Map Data Analyst

Jan 2022 – Dec 2022

- Applied linear, non-linear, and logistic regression models to over 100 quantitative evaluations of pedestrian routes, driving directions, search results, and auto-complete functions across iOS and macOS platforms, identifying correlations between user query behavior, geographic variables, and output accuracy.
- Diagnosed root causes of over 200 critical map data issues using statistical hypothesis testing and structured causal analysis, enabling faster, more consistent, and reproducible identification of systemic patterns and directly reducing reported navigation errors by 20%.
- Contributed to standardized analytical benchmarks and evaluation criteria, incorporating data scraping and automated extraction techniques, to model and improve map data reliability across four key dimensions: location relevance, name verification, address precision, and pin placement.

SKILLS

- Proficient in R [seurat], Python [pytorch, tensorflow, scanpy]. Basic knowledge of Java, SQL, CSS, HTML, MATLAB, Linux, C++, HPC/cloud computing, Git

PUBLICATIONS & ACKNOWLEDGEMENTS

- **Yong Jin Kweon** et al. (2025). Comparing Missing Data Imputation Methods for Patient-Reported Outcomes in Esophageal Cancer Research. <https://www.medrxiv.org/content/10.1101/2025.09.10.25335531v2>
- Samir S. Amin, **Jin Kweon**, Joshua S. Cheruvathur, Mehrnoush Dehghani, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Lorenzo Ferri, R. Trafford Crump (2025). Preoperative Quality-of-Life and Recovery Following Esophagectomy: Identifying Modifiable Factors Associated with Improvement. <https://www.medrxiv.org/content/10.1101/2025.07.30.25332378v1>
- **Yong Jin Kweon**, Chenyu Liu, Gregory Fonseca, Jun Ding (2025). DTractor enhances cell type deconvolution in spatial transcriptomics by integrating deep neural networks, transfer learning, and matrix factorization. <https://www.biorxiv.org/content/10.1101/2025.04.12.648541v1>
GitHub: <https://github.com/mcgilldinglab/DTractor>

- Chenyu Liu, **Yong Jin Kweon**, Jun Ding (2023). scBeacon: single-cell biomarker extraction via identifying paired cell clusters across biological conditions with contrastive siamese networks. <https://arxiv.org/abs/2311.02594>
GitHub: <https://github.com/mcgilldinglab/scBeacon>
- Yuying Lu, Camille Morencé, **Yong Jin Kweon**, Archer Yi Yang, Jun Ding (2025). SC2CD Co-optimizes Spatial Clustering and Cell Decomposition through Bidirectional Information Flow in Single-Cell Spatial Transcriptomics.
- Haber, J.R. (2020). Sorting Schools: A Computational Analysis of Charter School Identities and Stratification. Sociology of Education. <https://doi.org/10.1177/0038040720953218>

PRESENTATIONS

- Yousif Salman, Emad Mohammed, **Yong Jin Kweon**, Shayan Dhillon, Trafford Crump. (2026). Context-Aware Prompt Engineering for LLM-Based Missing Data Imputation in Patient-Reported Outcomes. IEEE Engineering in Medicine and Biology Society. Toronto, Canada. July 26-30, 2026.
- Xiuhui Yang, Jingtao Wang, **Yong Jin Kweon**, Chenyu Liu, Gregory Joseph Fonseca, Jun Ding. (2026). DTractor: Dual-reference modeling enables biologically coherent deconvolution of spatial transcriptomics. ISMB MLCBS. Washington, D.C., USA. July 12-16, 2026.
- **Jin Kweon**, Trafford Crump. (2025). Comparing Missing Data Imputation Methods for Patient-Reported Outcomes in Esophageal Cancer Research. Digital Health Symposium: From Research Infrastructure to Clinical Care. Montreal, Canada. Nov 4, 2025.
- Benjamin Tordjman, **Jin Kweon**, Nataliya Burbani El Dirani, Mehrnoush Dehghani, Mathieu Rousseau, Carmen Mueller, Sara Najmeh, Jonathan Spicer, Jonathan Cools-Lartigue, Lorenzo Ferri, Trafford Crump. (2025). Evaluation of textbook outcomes and long-term prognosis following esophagectomy: a retrospective cohort study. International Society for Disease of the Esophagus (ISDE) World Congress. Brisbane, Australia. Sep. 18-20, 2025.
- Nisha Suarez, **Jin Kweon**, Mehrnoush Dehghani, Lorenzo Ferri, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Jonathan Spicer, Trafford Crump. (2025). Testing the effect of changes in emotional distress on post-esophagectomy outcomes. European Society for Diseases of the Esophagus (ESDE) conference. Amsterdam, Netherlands, May. 12, 2025. DOI: 10.17605/[OSF.IO/UWBN2](https://osf.io/UWBN2)
- Samir Amin, Joshua Cheruvathur, **Jin Kweon**, Mehrnoush Dehghani, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Lorenzo Ferri, R. Trafford Crump. (2025). Rethinking Recovery Predictors in Esophageal Cancer Surgery. European Society for Diseases of the Esophagus (ESDE) conference. Amsterdam, Netherlands, May. 12, 2025. DOI: 10.17605/[OSF.IO/QNHSE](https://osf.io/QNHSE).
- Albouy, P., **Kweon, Y.J.**, Whittaker, H., Baillet, S., Zatorre, R.J. (2018). Enhancing learning-related plasticity with information-based neuromodulation. Auditory learning and plasticity symposium. Montreal, Canada, Oct. 23, 2018.